

Designing for Appreciation: How Digital Spaces Can Support Art and Culture

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ABSTRACT

Throughout history, a complex network of human actors shaped how the general public perceived art. Today, social media platforms and their algorithms influence artful experiences for billions. How has this changed the appreciation and perception of art? The role of visual art as items we use to define ourselves and our societies motivates research to explore how recommendation algorithms impact our ability to appreciate and perceive art. There are three aspects to explore: the art metadata and algorithm functionality, conversations with artworld experts, and redesigning digital art experiences. These three methods will follow open science practices and methods by releasing open-access datasets and research prototypes. Ultimately, this proposed thesis aims to contribute to theories of content base algorithmic recommendation and its role in presenting art and culture to users.

CCS CONCEPTS

• **Human-centered computing** → HCI theory, concepts and models.

KEYWORDS

datasets, recommendation algorithms, thesis proposal, art

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1 INTRODUCTION

The international art community is a social network of vast complexity and size as it connects artists with curators and audiences worldwide. As such, how this network functions has a profound social impact on the world. A notable part of the art community, the practice of art curation, recommendation and appraisal have become the next challenge for data science and computation. As art was created, humanity worked to maintain and display it, from religious rituals to wealthy patrons and private collections, then in public museums, and now with digital platforms [4]. Modern algorithmic recommendations mandate the curation and presentation

of art and information to the public. In this thesis, I am proposing an exploration of the impact of the current algorithmic curation on the public's ability to appreciate and perceive artistic and cultural experiences.

The proposed research will explore the existing art and culture recommendation algorithms, from advanced neural networks such as deep learning approaches to collaborative filtering. The work will compare the effectiveness of current algorithms to recommend visual art and consider the social and societal impact of the algorithms on cultural experiences. The methodology to explore this area of research requires the development of datasets of test art metadata. This analysis will also investigate the issues of fairness and the impact of the algorithms for recommending new artists, artists from minority groups, or non-Western cultures. Furthermore, I will develop prototypes with differing degrees of human agency to visualise various potential futures of algorithmic curation. Previous research argues that the underlying platforms for the recommendation and classification algorithms, such as social media and digital streaming services, are taking the place of cultural intermediaries of the past, such as curators, newspapers, radio, and television [29]. The impact of this shift towards 'infomediaries' on users' tastes, perceptions, and appreciation of art and culture has yet to be explored. My research aims to explore the role of these recommendation algorithms because of their potential to shape how we see the artefacts that fill our lives and define our culture.

The motivation for this thesis is grounded in research and theoretical exploration of technology as cause and effect in art and culture [10]. In this thesis, I will include references to music, film, and other cultural artefacts since many recommender algorithms applied to these domains have similar features and impacts on the public as those applied to visual art (paintings, sculpture, graphics, and prints).

The first motivating point for my research is that research in computation and art has been happening since the 1950s with attempts to identify variables that describe art and artefacts [23, 26, 28, 36]. Furthermore, most research exploring the relationship between recommendation algorithms and art has focused primarily on music and audio streaming [13, 14, 31]. Research beyond music has been explored over the last few years using advances in machine learning and neural nets but does not explore art recommendation algorithms. Instead, the research has led to the ability to classify, identify and appraise art with computer vision techniques and has also led to the current explosion in generative AI works [33]. I argue there must be more research exploring the impact of recommendation systems on human experiences with art and culture. Therefore, this thesis investigates the diverse impact of these systems in the context of art appreciation.

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The second motivating point is that the lens of Human-Computer Interaction (HCI) applies well to researching algorithmic curation's impact on art and culture. Art and HCI are by no means strangers to collaboration; years of work by Bardzell and Bardzell, Blythe et al., and Andersen et al. have explored questions related to how digital media have changed the creation and presentation of art [3, 7, 11]. Focusing on user experiences with interactive technology that targets audiences interested in art, HCI offers a variety of methods applicable to this research (see Methods). Today's discourse around digital platforms and the role of data in our lives applies to all digital content experienced by billions [37]. As society wrestles with its own biases that are multiplied and widespread due to the inherent power of social media and technology, the questions of human agency and values are amplified [6, 16, 22]. This thesis will bridge current discussions on the power of algorithms with historical discussions on the power of art and artefacts. By exploring users' experiences, new interfaces for human agency and cultural values can be proposed that facilitate critical engagement and careful curation. This thesis will involve multi-faceted methods grounded in existing HCI research. The following section presents the three primary research paths contributing to the overarching thesis.

2 METHODS

While this thesis will formally be presented as one coherent body of work composed of multiple supporting parts, this section is separated into the primary research efforts for ease of discussion. Each research effort will be explored through various papers and projects. The three areas are as follows:

- (1) What factors, values or variables are currently at play in algorithmic curation? How do the algorithms work?
- (2) What do experts and regular users have to say about the impact of recommendation algorithms on their engagement with art and culture?
- (3) What does the future of appreciating art look like if algorithmic curation remains unchanged? What if it changes?

2.1 Art Data and the State of Algorithmic Curation

As seen in recommendation algorithms used on platforms such as social media and streaming services, algorithmic curation is built to promote and optimise certain factors. These factors are often the business values of the parent corporation [12, 17, 22, 35]. Most of these algorithms are engagement-based, providing users with content similar to the content they had previously engaged with. Understanding the current state of algorithmic curation across social media and other platforms requires cross-examining the popular systems and their impact on users.

Completed Research: A single shared test dataset was developed for the comparative analysis of multiple algorithms. The research began by gathering 30,000 visual artworks from major auction house websites. Auction house websites proved one of the most reliable sources of well-formatted art data. The data was labelled with 22 variables of interest that can be used to explore and classify the artworks. A selection of over 10,000 lots was cleaned and manually verified for release as a public dataset for training, validation,

and testing. Exploring the usefulness of this novel dataset for the task of art recommendation requires the establishment of a benchmark to which to compare. The work by Abera Yilma et al. offered an approach involving a Latent Dirichlet Allocation (LDA) topic model as the basis for visual art recommendation [1]. Following the methodology, I ran the test dataset using the same LDA model. Not only was our analysis of the larger dataset able to provide a higher coherence score across various topic distributions, but the data also provided multiple distinct topics differentiated by the type of art within the auction lots (currently under review). The test dataset and trained topic model serve as the groundwork for additional work comparing algorithmic recommendation and curation systems for art. Other research has been done using natural language processing (NLP) systems and neural networks in processing art data [19, 32]. Ultimately, understanding the impact of how different algorithms present art to users will require a full comparison.

Future Research: First, expanding the dataset to include all 30,000 cleaned and manually verified auction lots will prepare it for the larger comparative research into different curation algorithms. This proposed research will analyse and compare the developed algorithms within this field, ranging from small-scale models to the more obscure recommendation algorithms used by major social media platforms. Should one of the algorithms be deemed the most impactful at shaping people's experience of art, it could result in dramatically different outcomes for how society interacts with art and culture. Therefore, it is also necessary to consider the examined algorithms' potential ethical impact. I propose using a methodology that involves developing a scoring system and looking at approaches that can moderate the effect of relying on a single approach, such as Bayesian model averaging [25]. Algorithm comparison requires carefully preparing the available algorithms trained on the test dataset. To ensure the quality of the comparative analysis, the quality of the test datasets will need to be reviewed by art experts. Expert verification of the dataset will combine the user research outlined in the following section and integration with digitised museum collection APIs such as those from the Harvard Art Museum [38].

The accuracy, efficiency, and fairness of curating artefacts are the primary comparison metrics of interest. Typical accuracy and efficiency metrics will include a range of standard performance metrics (log-loss, average accuracy, number of parameters, area under the curve (AUC), convergence rate, root mean square error (RMSE), and precision-recall) [15, 24, 27, 39]. Typical fairness metrics will include the analysis of the algorithms' recommendations for sub-groups (such as artefacts developed by minority artists, or from non-Western cultures). These metrics will inform the algorithm selection and algorithm averaging process.

Risks and Considerations: There are two primary risks involved with the proposed methodology. First is the challenge of replicating the various algorithms accurately. Some of the most popular recommender systems are proprietary and difficult to replicate. Second, there may be a need for additional data collection. The need for more data poses an ethical consideration: As the dataset grows or is segmented differently to focus on different art, cultures, or artists, more potentially sensitive information is needed. For example, questions regarding the gender, sexual, or racial identities of artists or

other stakeholders will require data that is not currently found in the datasets. It is vital to consider the risks the research poses as it progresses. Understanding the dangers and challenges will better prepare us to identify the appropriate resources to handle the research responsibly.

2.2 Experiencing the Impact of Algorithmic Curation

The second pillar of this thesis rests upon the lived experiences of experts, artists, and the public. Experts and artists have made up the fabric of the art world for quite some time; their roles have also changed drastically even before the introduction of algorithmically driven platforms. On the other side of the exhibition wall stands the public, who has encountered the trends and seen the changes. All three groups are the users whose appreciation and perception of art and culture are the focal point of this thesis. Therefore, I will pursue a variety of human-centred research methods to uncover and process the users' perspectives on art and algorithmic curation.

Completed Research: I established a network of approximately 20 artists and experts for this topic. The network includes academic curators, private gallery curators, and artists of varying degrees of fame and financial success. I interviewed this network and analysed their responses using affinity clustering and thematic analysis. Additionally, I supported the findings from this network with almost 40 survey responses from the public. The interviews and the surveys asked the participants about their taste in art, the public's perception of art, and what they would change to improve the internet or social media for art experiences.

Based on the responses from our participants, I presented four features essential to our participants for the appreciation and perception of art and culture. These four were depth, conversation, connection, and time (currently under revision). Depth refers to additional information, metadata, and facts accurate to the artefact. Conversation is the ability to have subjective thoughts and opinions from anyone openly discussed aside from the facts of the depth pillar. Connection calls for clarity around sources of information and cultural associations that tie one piece to other works or items worldwide. Finally, time refers to sitting with the artefact and processing the information and the content of the previous three pillars. Additionally, I worked to contextualise the four features within the existing literature on art theory [8]. The participants noted that the theories of Expressionism and Instrumentalism are still highly relevant when appreciating art. Participants argued that good art sparks some sort of emotional (Expressionism) or intellectual (Instrumentalism) reaction within the audience. Our participants argued that through the rise of digital platforms, there is a newly added focus on an artwork's transactional value or marketability. For more information on the art theories considered in the study, see [8, 9, 20, 21].

Future Research: I plan to reconvene the network of experts and artists to explore further research topics. Specifically, co-design workshops and participatory design will explicate the thoughts and ideas of the experts into design concepts that can be visualised. Through these exercises, I will explore the future of art on digital platforms and what physical spaces within museums and galleries

may look like to reflect the growing presence of digital life and culture. The motivation for this work is that the original development of most algorithmic curation has yet to be explicit or intentional about the involvement of experts and creators. Our convened network's expertise can inspire the future design and development of art and cultural spaces, both digital and physical. Similarly, by maintaining, possibly expanding, this group of experts and artists, I will also be able to present them with the prototypes from the third part of the thesis, as described below. They can serve as an expert audience that can validate and effectively critique the prototypes as I iterate on potential futures of art and culture in digital spaces. Furthermore, I will rely on the public to serve as user testers for the various designs and prototypes developed as part of the thesis work.

Risks and Considerations: Research with a network of experts, artists, and, occasionally, the public is human subjects research. Therefore, it needs to consider the impact of the methodology on the participants. I must ensure the study design is thoroughly considered and reviewed by the appropriate research ethics committees. The study reviews will also allow other established researchers to consider and evaluate the methods. Ideally, their consideration and approval will alleviate the risk that these methods are ineffective for the insights I am searching for. Simultaneously I must be aware that the network of experts and artists is a small and unique sample. The ability to draw generalisable findings will be limited. In the past, it was shown that snowball sampling could grow the number of artists and experts in the network. Thus I must work to ensure I get appropriate involvement from the larger public as participants. While I may continue to face the challenge of small sample sizes, by increasing the variety of backgrounds and perspectives, the research will be better contextualised in various experiences.

2.3 Developing the Future of Algorithmic Curation

The final pillar of the proposed thesis work involves designing, developing, and refining prototypes that will envision various future experiences of art and culture experiences in a digital world. Within this portion of the proposed work, it becomes clear that each of the pillars of the thesis is dependent on the others. The previous two provide insights into algorithmic curation data and human experiences, while this section focuses on creating artefacts for a better and fairer experience of art. Again I will rely on the literature on human-centred methods for designing and iterating through the potential futures of algorithmic curation of art and culture experiences.

Completed Research: So far, my research efforts have focused on grounding the first two areas of work. That work has created a baseline amount of data and feedback from which the greater work of the thesis can begin. Therefore, the completed research for this thesis portion consists primarily of study designs and drafts for potential prototypes.

Future Research: I envision one study exploring human agency and content-based decision-making in algorithmic curation. The first iteration will be a "Wizard of Oz" study where two participants with some previous relationship are paired off. Wizard of Oz studies

are a common type of HCI method where the functionality of a digital system is simulated either by a researcher or by one of the participants themselves [18, 30]. Please refer to [30, 34] for additional information regarding the potential methods available. One acts as the “selector” while the other is the “receiver”. The “selector” will be presented with a controlled selection of artefacts, likely from a data source such as Google Art and Culture, and is asked to select a sample of artefacts for their companion, the “receiver”. For each selection, the “selector” will be asked why they chose the piece and how they want to present it to the “receiver”. The “receiver” will then be presented with the curated selection of works, and I will ask them about their thoughts and experiences. Finally, the two will switch roles and repeat the experiment. For iterations and variations, I can explore how much control either the “selector” or “receiver” has, and I can even alter the presentation experience. Furthermore, what might start as an in-person Wizard of Oz study can be moved to a fully digital interface to understand better how that setting might alter participants’ experiences. Each prototype or experience will be reviewed iteratively through user testing and expert review. Before the study described above is deployed, I would survey the experts and artists about the type of selection and presentation process. Their feedback before and after the study will work to validate or challenge assumptions and insights from the research resulting in better subsequent designs.

Risks and Considerations: This work will result in prototypes, designs, and outputs that are research artefacts. I am not following a move-fast-break-things approach. As I share the work with industry professionals in and within the art world, I need to manage expectations for what these artefacts, and digital prototypes can become. Again user testing and involving participants with any artefact requires appropriate review and consideration by the relevant ethical review committees for each applicable study.

3 CONTRIBUTIONS

A variety of potential contributions may arise as the research of the three major areas of the thesis is synthesised. For ease of discussion, these contributions of this thesis can be grouped into methodological, theoretical, and practical categories. Similarly to the methods described above, these contributions will be interwoven to ensure they are rigorously applied and represent best practices in the field of HCI.

Methodological Contributions: Previous HCI methodology, especially in educational communication research, has seen researchers enter the museum or gallery space to explore different ways to convey information to an audience [2]. The co-design workshops I propose will bring these art world experts out of the physical world and consider how audiences encounter art and culture in the digital world. I will draw on speculative and human-centred design principles to explore novel ways users think about art and culture [5]. My approach to using data for content-based algorithmic curation will also present novel ways of considering how algorithm-driven platforms operate. Furthermore, this thesis will follow Open Research principles, a novel approach to sharing (art) datasets and models. Using tools such as the Open Science Framework and the Open Data Institute as inspiration, this thesis can provide the first

examples of sharing art metadata for other academics to access and refine. Any models or recommendations and curation methods arising from this approach will also bridge the value of the art world across digital platforms and corporations.

Theoretical Contributions: The concept of art theory in the modern days is evolving. By understanding how much a focus on art appraisal has influenced art appreciation through the digital world, it can be understood what values are used to define the role of artefacts in society today. The thesis aims to reimagine how art and culture are displayed and how users perceive the artefacts. This will contribute to reimagining the theoretical assumptions used in building current social media and how they present information (engagement vs content-based curation). While these insights will not solve or dictate how people should experience art and culture, they will question the status quo and present theories on how algorithms can better serve humanity concerning its artefacts. Finally, based on the feedback from the exploratory study, the building blocks of a theoretical framework emerge that can be used to design and develop digital artistic and cultural spaces. Understanding how depth, conversation, connection, and time can serve as four aspects of appreciating and experiencing art and culture will be refined and validated through the user research conducted during the thesis work. Through this research, new questions will arise about what philosophies and value structures exist for art and culture and how much of the changes can be assigned to the role of digital platforms.

Practical Contributions: Considering a large portion of the proposed thesis will focus on creating various prototypes and designs, there will be several practical contributions in the form of artefacts that can be documented and shared with the research community. One contribution that already exists is the current version of the art auction dataset. This dataset has been released publicly and will be expanded throughout the thesis. I have already explored the data using an initial LDA topic model that identifies distinct topics underlying the data describing art auction lots. The initial LDA model results and the code will be documented and released. Like the dataset, the model must be updated, retrained, and validated on any data updates. Finally, the items created from the co-design workshops and iterative design process that will envision various future examples of algorithmic curation will be the last and largest set of practical contributions. These academic research prototypes can be shared and distributed among researchers interested in exploring questions regarding algorithmic curation and how participants interact.

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